

# Topological Memory.

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# Introduction

Today:

- ▶ Proof.
  1. Walkthrough the math

# Our Code

## Code

In this session, we consider LDPC codes, where each bit adjoins to  $\Delta_1$  checks, and each check adjoins to  $\Delta_2$  bits. We denote by  $\Delta = \Delta_1 \Delta_2$ .

# Local Decoder

## definition

We say that  $D$  is a local decoder if when applied against  $2p(1-p)$ -depolarized channel:

- ▶ There exists function  $M : (0, 1) \rightarrow (0, 1)$  such that the probability of each bit being flipped is  $M(2p(1-p)) < p^{\frac{100}{99}}$ .
- ▶ There is a constant  $l$  such that for any bit, there are only  $(\Delta)^l$  bits on which the event of it being flipped depends.

For example, if we take  $k$  large enough and consider the  $(n, k)$ -Toric, then both the swift-rule and the flip upon majority are local decoders.

# Storing Logical State - Classical Statement

## Logical State

For a random variable  $X$  we denote by:

1.  $\mathbf{E}_{\sim p}[X]$  - its expectation when measured after a single application of the  $p$ -depolarizing channel.
2.  $\mathbf{E}_{\sim q^{(t)}}[X]$  - its expectation when measured after  $t$  rounds, each initiated by a  $p$ -depolarized channel followed by an application of local decoder  $D$ .

# The Theorem - Classical Statement

## Theorem

There is  $p_{\text{th}} \in (0, 1)$  such that for any  $p < p_{\text{th}}$  and any bounded function  $f$  with a supremum  $|f|$  it holds:

$$\mathbf{E}_{\sim q(t)}[f] \leq \mathbf{E}_{\sim p}[f] + t|f|e^{-\Theta(\sqrt{n})}$$

## Lemma

There exists a constant  $c > 0$ , such:

$$\mathbf{E}_{\sim 2p(1-p)}[f \circ D] \leq \mathbf{E}_{\sim p}[f] + |f|e^{-c\sqrt{n}}$$

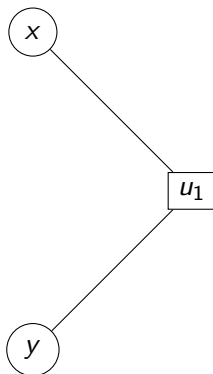
# Theorem Proof

By induction on  $t$ .

$$\begin{aligned}\mathbf{E}_{\sim q^{(t)}}[f] &= \mathbf{E}_{\sim q^{(t-1)}}[\mathbf{E}_{\sim p}[f \circ D]] \\ &\leq \mathbf{E}_{\sim p}[\mathbf{E}_{\sim p}[f \circ D]] + (t-1)|\mathbf{E}_p[f \circ D]|e^{-c\sqrt{n}} \\ &\leq \mathbf{E}_{\sim 2p(1-p)}[f \circ D] + (t-1)|f|e^{-c\sqrt{n}} \\ &\leq \mathbf{E}_{\sim p}[f] + t|f|e^{-c\sqrt{n}}\end{aligned}$$

When in the last transition we used the lemma.

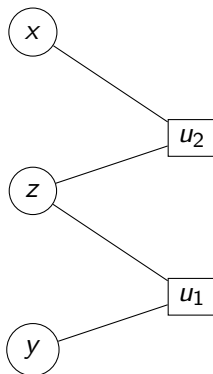
## Dependence Type I



$$\begin{aligned} & \mathbf{Pr}[(x, y) \in e | u_1 \text{ is satisfied}] \\ &= \mathbf{Pr}[x \in e | u_1 \text{ is satisfied}] \mathbf{Pr}[y \in e | u_1 \text{ is satisfied}] \\ &\leq \mathbf{Pr}[x \in e] \mathbf{Pr}[y \in e] \end{aligned}$$



## Dependence Type II



$$\begin{aligned} & \mathbf{Pr}[(x, y) \in e | z \text{ is not flipped}] \\ &= \mathbf{Pr}[x \in e | z \text{ is not flipped}] \mathbf{Pr}[y \in e | z \text{ is not flipped}] \\ &\leq \mathbf{Pr}[x \in e] \mathbf{Pr}[y \in e] \end{aligned}$$

## Definition

Consider the moment after the application of the depolarizing channel, and right before the correction by the decoder.

For a subset of qubits  $S$ , we say that a qubit  $x \in S$  is depended if there exists  $y \in S$  such:

1. Either  $x$  and  $y$  share a check that is not satisfied. (Type I)
2. Or,  $x$  and  $y$  have checks which share a bit that was flipped by the application of the channel. (Type II)

# Dependence

## Claim

Consider the application of a  $2p(1 - p)$ -depolarizing channel. Let  $S$  be a subset of qubits chosen uniformly random<sup>1</sup>, then the probability that there are more than  $\frac{1}{100}$  depended qubits in  $S$  is less than  $2e^{-c\sqrt{n}}$ .

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<sup>1</sup>each qubit is taken with probability  $\frac{1}{2}$

# Dependence

## Proof

First, for an arbitrary  $S$ , denote by  $J$  the qubits that depend on qubits in  $S$ . Then, the expected number of depended qubits in  $S$  is:

$$\begin{aligned} & \mathbf{E}_{\sim 2p(1-p)} [\text{Type II depended qubits in } S] \\ & \leq \mathbf{E}_{\sim 2p(1-p)} [|J|] \leq 2p(1-p)\Delta' |S| \end{aligned}$$

Similarly, let  $J'$  be the qubits in  $S$  that connected to unsatisfied check, then:

$$\begin{aligned} & \mathbf{E}_{\sim 2p(1-p)} [\text{Type I depended qubits in } S] \\ & \leq \mathbf{E}_{\sim 2p(1-p)} [|J'|] \leq \frac{1 - (1 - 2p)^{\Delta_2}}{2} \Delta' |S| \end{aligned}$$

# Dependence

We are going to use the following inequality:

**Theorem 1** (McDiarmid's inequality). *Let  $X_1, \dots, X_n$  be independent random variables, where  $X_i$  has range  $\mathcal{X}_i$ . Let  $f: \mathcal{X}_1 \times \dots \times \mathcal{X}_n \rightarrow \mathbb{R}$  be any function with the  $(c_1, \dots, c_n)$ -bounded differences property: for every  $i = 1, \dots, n$  and every  $(x_1, \dots, x_n), (x'_1, \dots, x'_n) \in \mathcal{X}_1 \times \dots \times \mathcal{X}_n$  that differ only in the  $i$ -th coordinate ( $x_j = x'_j$  for all  $j \neq i$ ), we have*

$$|f(x_1, \dots, x_n) - f(x'_1, \dots, x'_n)| \leq c_i.$$

For any  $t > 0$ ,

$$\Pr(f(X_1, \dots, X_n) - \mathbb{E}[f(X_1, \dots, X_n)] \geq t) \leq \exp\left(-\frac{2t^2}{\sum_{i=1}^n c_i^2}\right).$$

# Dependence

Thus, the probability that the number of depended qubits in  $S$  is  $|S|^{\frac{3}{4}}$  far from the expectation:

$$\Pr_{\sim 2p(1-p)} \left[ \# - \mathbf{E}[\#] \geq |S|^{\frac{3}{4}} \right] \leq e^{-2 \frac{|S|^{\frac{3}{2}}}{n \Delta^I}}$$

So, for any  $|S| > \alpha n$  we have:

$$\Pr_{\sim 2p(1-p)} \left[ \# - \mathbf{E}[\#] \geq |S|^{\frac{3}{4}} \right] \leq e^{-\frac{2\alpha^{\frac{3}{2}}}{\Delta^I} \sqrt{n}}$$

# Dependence

Thus for small enough  $p \in (0, 1)$ , with probability greater than  $1 - 2e^{-\frac{2\alpha^{\frac{2}{3}}}{\Delta'}\sqrt{n}}$ , the number of depended bits is less than:

$$\left(2p(1-p) + \frac{1 - (1-2p)^{\Delta_2}}{2}\right) \Delta' |S| + |S|^{\frac{3}{4}} \rightarrow \frac{|S|}{100}$$

## Lemma Proof

Denote by  $S$  the subset of the flipped qubits, and denote by  $de(S)$  the number of depended qubits in  $S$ .

$$\begin{aligned}\mathbf{E}_{\sim 2p(1-p)}[f \circ D] &= \sum f \circ D(S) \cdot \mathbf{Pr}[S] \\ &\leq \sum_{\substack{de(S) < \frac{1}{100}|S| \\ |S| \geq \alpha n}} f \circ D(S) \cdot \mathbf{Pr}[S] + 2|f|e^{-c\sqrt{n}} \\ &\leq \sum_{\substack{de(S) < \frac{1}{100}|S| \\ |S| \geq \alpha n}} f(S)M(2p(1-p))^{\frac{99}{100}|S|} + 2|f|e^{-c\sqrt{n}}\end{aligned}$$



## Lemma Proof

$$\begin{aligned} &\leq \sum_{\substack{de(S) < \frac{1}{100}|S| \\ |S| \geq \alpha n}} f(S)p^{|S|} + 2|f|e^{-c\sqrt{n}} \\ &\leq \sum_S f(S)p^{|S|} + 2|f|e^{-c\sqrt{n}} \\ &\leq \mathbf{E}_{\sim p}[f(S)] + 2|f|e^{-c\sqrt{n}} \end{aligned}$$

# Storing Logical State - Quantum Statement

## Logical State

Let  $\rho^{(0)}$  be a mixed state over codewords. Define:

$$\rho^{(t)} = D \circ \mathcal{N}_p^{(t-1)}$$

1.  $\mathbf{E}_{\sim p}[X]$  - its expectation when measured after a single application of the  $p$ -depolarizing channel.
2.  $\mathbf{E}_{\sim q^{(t)}}[X]$  - its expectation when measured after  $t$  rounds, each initiated by a  $p$ -depolarized channel followed by an application of local decoder  $D$ .

# The Theorem - Quantum Statement

## Theorem

There is  $p_{\text{th}} \in (0, 1)$  such that for any  $p < p_{\text{th}}$  and any bounded Hermitian operator  $X$  we have:

$$\mathbf{Tr} X\rho^{(t)} \leq \mathbf{Tr} X\mathcal{N}_p(\rho) + t\|X\|e^{-\Theta(\sqrt{n})}$$

## Lemma

There exists a constant  $c > 0$ , such:

$$\mathbf{Tr} XD \circ \mathcal{N}_{2p(1-p)}(\rho) \leq \mathbf{Tr} X\mathcal{N}_p(\rho) + 2\|X\|e^{-\Theta(\sqrt{n})}$$

